

DBCLS BioHackathon 2026 report: A knowledge base for the Marchantia genome database MarpolBase

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Introduction

The liverwort *Marchantia polymorpha* (Bowman et al., 2017) has drawn attention as a new-generation model plant in developmental and evolutionary biology, and a high-quality reference genome has been determined for it (Tanizawa et al., 2025). For bioinformatics the species is attractive for a further reason: the community maintains an agreed gene nomenclature and gene identifier system, and researchers use that system when they describe genes in their papers. A gene mentioned in the literature can therefore be resolved to an identifier with far less ambiguity than in most species, which is an advantage both for extracting information from papers and for integrating data across sources.

MarpolBase, the genome database for the species, was published in *Plant and Cell Physiology* (Tanizawa et al., 2025), selected as an Editor's Choice, and accompanied by a commentary placing it among the community-building resources that make *Marchantia* an accessible model system (Boxall & Haseloff, 2026).

Our goal at the DBCLS BioHackathon 2026 was to turn that database into a knowledge base: to publish its gene, expression and literature data as RDF, to expose them through a SPARQL endpoint, and to make them reachable from AI agents through a Model Context Protocol (MCP) server.

The question behind this was practical. *M. polymorpha* is not indexed by the major cross-reference hubs, so it is not obvious that a species-specific resource can be linked to the wider set of life-science databases at all. Much of what follows is therefore about *where* the join actually happens.

Table 1 lists what was produced.

Table 1: Resources and tools developed or extended during BioHackathon 2026.

Name	Description and URL
MarpolBase RDF	RDF distribution of the MarpolBase gene, expression and literature data. https://marchantia.info/rdf/
MarpolBase SPARQL endpoint	Public endpoint, registered with TogoMCP and RDF Portal. https://marchantia.info/sparql
MarpolBase MCP server	14 tools giving agents typed access to genes, expression, co-expression and literature. https://marchantia.info/mcp

MarpolBase RDF and MCP server

What was published

The gene, expression and literature data served by MarpolBase were converted to RDF and released. The SPARQL endpoint was registered with both TogoMCP (Kinjo et al., 2026) and RDF Portal, so MarpolBase can now be reached from the cross-database query tooling maintained in Japan. On top of the endpoint we built an MCP server with 14 tools, which lets an LLM agent query the resource without writing SPARQL by hand.

The tools fall into five groups: gene lookup and search, expression profiles and condition comparison, co-expression networks, curated gene–literature links, and controlled-vocabulary resolution. A `run_sparql` tool and a `describe_schema` tool are also provided so that an agent can fall back to raw SPARQL when the typed tools are not enough.

The literature layer is the part the gene identifier system pays off in, and it is described in its own section below. Both the gene–paper links and the LLM-drafted function summaries are in the RDF, so an agent retrieves them as data rather than re-reading the papers.

Linking the literature to the genes

The gene–literature layer was assembled in a separate effort and imported into MarpolBase. The pipeline has four steps:

1. Reference lists were parsed out of PDFs and each paper’s DOI, PubMed ID and PMC ID completed against CrossRef and the NCBI E-utilities. **1,399 papers** were surveyed.
2. The full text of each paper — the abstract only where no full text could be obtained — was read and the *Marchantia* genes it mentions identified. 3,020 of the resulting records come from full text, 88 from full text plus supplementary files, and only 39 from an abstract alone.
3. Each (gene, paper) pair was tagged on two axes: an **evidence** code (experimental, sequence or citation) and a **role** (subject, tool, comparator or background), together with a two-to-three-line function note taken from the text. A pair that is both experimental and subject is a **core reference** — the paper actually did experiments on that gene — and a gene page shows those by default, with the peripheral mentions behind a “more” control.
4. Gene identifiers were normalised to the MpTak_v7.1 assembly and per-gene JSON was generated for MarpolBase to ingest.

The result: **510 of the papers carry at least one gene link**, giving **3,147 gene–paper records** that name **1,455 distinct gene symbols and identifiers**. 1,105 of those resolve to a v7.1 gene ID, which is 85% of the records; 1,000 records over 530 genes are core references.

Why the linking was feasible at all

This is where the point made in the introduction becomes concrete. The exercise works because the *Marchantia* community has a **gene nomenclature convention and a stable gene ID system, and both have been adopted widely enough that authors use them in their papers**. A gene mentioned in running text is therefore not a free-form string: MpBNB, MpPAL1 and Mp3g23300 belong to a registered vocabulary, so a mention can be matched against the nomenclature table and carried to an identifier.

Two details show how much of the work that convention absorbs:

- **Author-specific spellings still resolve.** A paper writing MpCYCD;1 is matched through the nomenclature entry that carries both that symbol and the identifier Mp8g17230, so a naming style local to one paper does not break the link.

- **Older papers keep their links.** A correspondence table from earlier assemblies to MpTak_v7.1 lets papers that predate the current assembly resolve to current identifiers — which is what makes a survey reaching back to the 1970s worth doing at all.

Without that discipline the same reading effort would have produced ambiguous strings rather than links, and there would have been nothing to put in the RDF. The asymmetry is worth stating plainly: the curation was affordable here because the naming was settled first.

Use cases

We exercised the server with three analyses driven entirely from natural-language prompts.

In the first (Figure 1), the prompt “tell me the conditions in which *MpBNB* is highly expressed” was answered by retrieving the mean TPM of *MpBNB* across all 161 conditions in MarpolBase, together with the condition attributes (tissue, accession, stage, sex, mutant, treatment, infection). The gene reaches 27.8 TPM in the antheridium and stays near zero in thallus, gemma and spore, which the agent could report because the expression values carry condition metadata rather than bare sample identifiers.

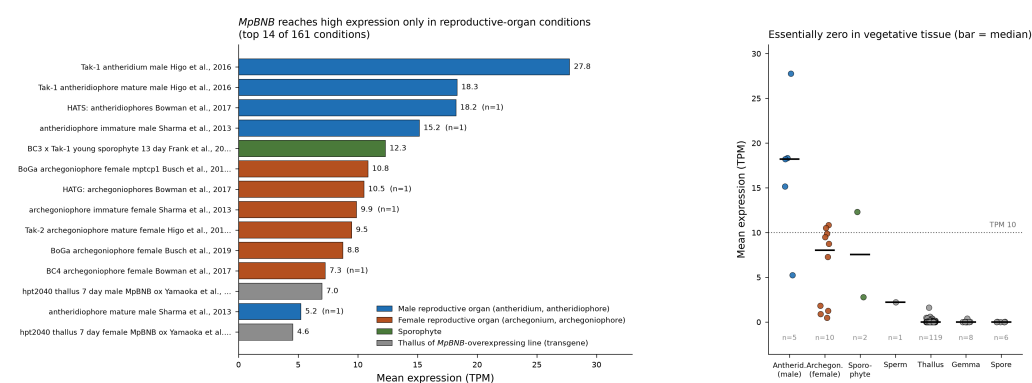
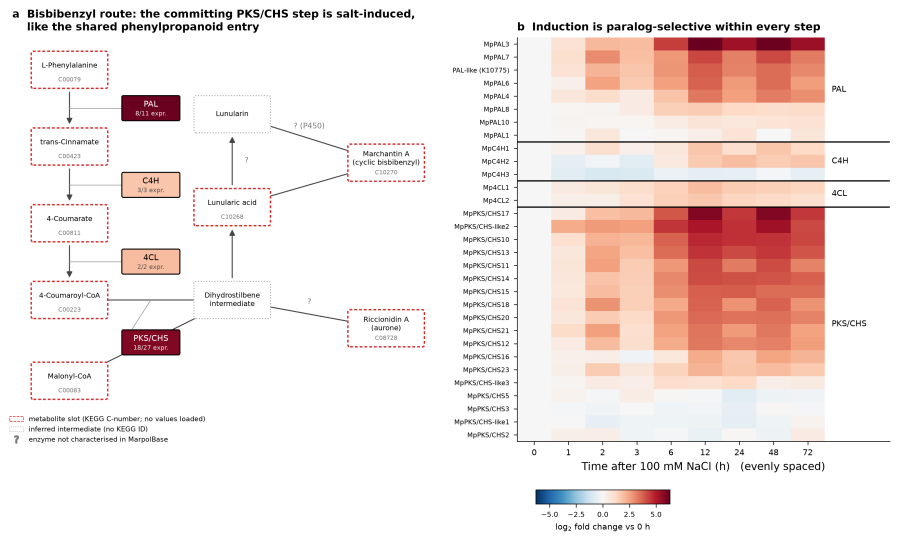


Figure 1: Expression of *MpBNB* across the 161 conditions in MarpolBase, retrieved through the MCP server from a single natural-language prompt. Left: top 14 conditions by mean TPM, coloured by organ type. Right: the same values grouped by tissue.

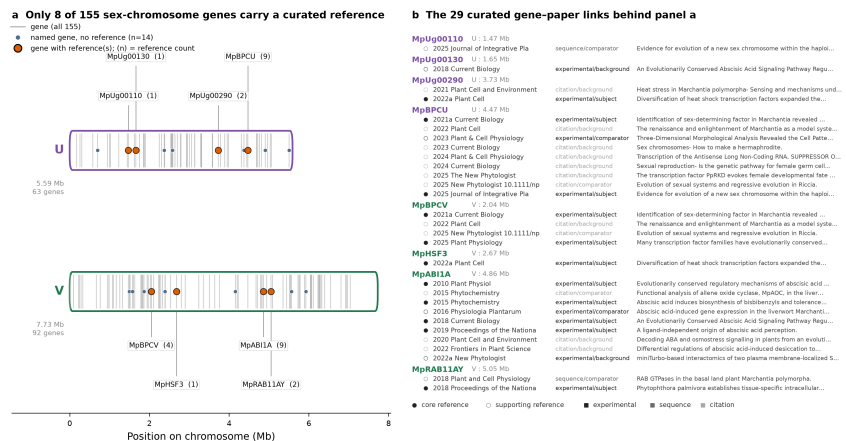
In the second (Figure 2), we followed the bisbibenzyl biosynthesis route under salt stress. Expression values came from MarpolBase; the pathway context — compound identifiers, reactions and enzyme references — came from external resources reached through TogoMCP. The committing PKS/CHS step and the shared phenylpropanoid entry are both salt-induced, and the induction is paralog-selective within every step.



Panel a: enzyme boxes coloured by the strongest induction among that step's expressed paralogs; 'n/n expr' = paralogs above the expression floor / all paralogs annotated in MarpolBase. Panel b: rows are individual genes (log₂ fold change vs 0 h), grouped by step and sorted by 12 h response. 31 of 43 genes shown; 12 omitted — 11 below the 1 TPM floor (a fold change from a ~0.01 TPM baseline is not interpretable) and MpPAL5 (Mpolg14170), whose profile MarpolBase returned an error for and was therefore not assessed, not treated as absent. Steps from lunularic acid onward have no characterised enzyme in MarpolBase and are left uncoloured. Reaction topology is a literature schematic, NOT from KEGG — lunularic acid (C10268), marchantin A (C10270) and riccinonin A (C08728) exist in KEGG COMPOUND but are placed on no KEGG pathway map. Expression: Tai-1 thallus 7 d, Wu et al., 2021 (n_runs 3-7 per time point).

Figure 2: The bisbibenzyl route under salt stress. (a) Pathway schematic with enzyme steps coloured by the strongest induction among the expressed paralogs of that step. (b) log₂ fold change of individual genes over a 100 mM NaCl time course, grouped by step.

The third (Figure 3) asked for the genes on the sex chromosomes that are mentioned in the literature, drawn on the chromosomes. This one joins three layers at once — gene positions from the FALDO annotation, the curated gene-paper links, and the evidence and role codes on each link — and it is also a readable audit of the curation: of the 155 genes on chrU and chrV, 8 carry a curated reference, through 29 links over 23 distinct papers. An absence in the figure means no curated link, not an absence of literature, which is the kind of distinction a resource has to make explicit if an agent is to reason over it.



MpTbk v7.1. Gene positions and chromosome lengths from MarpolBase FALDO annotation (chrU 5.59 Mb / 63 genes; chrV 7.73 Mb / 92 genes). Two further genes carrying MpUgMpolg identifiers (Mpolg1228, Mpolg0855) have no location record and are therefore absent from panel a — not assessed, not absent from the chromosome. Panel b lists every reference MarpolBase links to these genes (29 links over 23 unique papers; filled circle = mpolgConReference, 'evidence/role' are MarpolBase's own curation codes. Absence of a reference means no curated link, not absence of literature.

Figure 3: Genes on the *M. polymorpha* sex chromosomes that carry a curated reference, assembled from one prompt. (a) chrU and chrV with all 155 genes, the named ones marked, and the 8 with a curated reference highlighted. (b) The 29 gene-paper links behind panel a, with MarpolBase's own evidence and role codes.

What MarpolBase and TogoMCP each contribute

Working through these analyses made the division of labour between a species-specific resource and a cross-database hub concrete (Table 2).

Only MarpolBase can supply measured expression across its 161 conditions, the co-expression network (PCC / HRR / MR), DESeq2 differential expression between two conditions, gene–literature links curated with evidence and role, and controlled-vocabulary resolution with PECO and PO identifiers. It also carries its own KEGG Orthology assignments — `mpo:hasKEGG` for 6,563 genes. This last point mattered in practice: *M. polymorpha* is absent from KEGG GENES, and the enzyme nodes in Figure 2 could be coloured **only** because MarpolBase assigns KO numbers itself.

Only TogoMCP can convert identifiers across 119 databases (TogoID) (Ikeda et al., 2022), search the species-agnostic references (UniProt, PDB, Reactome, Rhea, ChEMBL, MeSH), reach the NCBI E-utilities for SRA, GEO and PubMed, and dispatch SPARQL to a choice of endpoints.

Table 2: Where the two resources overlap, and how the overlap differs in kind.

Capability	MarpolBase	TogoMCP
SPARQL	One own graph (GRAPH must be given explicitly)	A choice of endpoints
Literature	<i>Marchantia</i> papers linked to genes, with evidence and role	Generic PubMed / MeSH search
Compounds	Not covered	ChEBI / Rhea / PubChem identifier conversion
Gene expression	Measured values for 161 conditions	Not held

The two do not meet at the gene identifier

The most useful negative result of the project: **TogoID holds no *M. polymorpha* gene dataset, and MarpolBase's `get_gene` does not return UniProt accessions.** There is therefore no identifier-conversion path that carries Mp3g23300 directly into the TogoMCP side. We also confirmed that `pfam → uniprot` does not exist as a TogoID route (404), while `rhea/chebi`, `chebi/pubchem_compound`, `uniprot/rhea`, `uniprot/pdb` and `go/uniprot` do (in both directions).

The two resources meet instead at the level of **shared vocabularies**:

1. KO numbers — the junction that made the two-layer map in Figure 2 possible.
2. GO and Pfam — assigned by MarpolBase, dereferenceable on the TogoMCP side.
3. Compound identifiers (ChEBI / PubChem) — the junction for metabolome data.

Going through UniProt instead means searching by protein name and organism, which for paralog families such as MpPAL1–10 is many-to-many and therefore unreliable.

The practical rule we arrived at: use MarpolBase for measured values, conditions, co-expression and curated literature in *M. polymorpha*; use TogoMCP to carry those results outward to structures, reactions, compounds, orthologs in other species and sequence archives; and bridge the two with KO, GO and compound identifiers rather than with gene identifiers.

Connecting to the server

The MCP server is a remote, streamable-HTTP endpoint at <https://marchantia.info/mcp>. It needs no local installation: a user adds the URL as a connector in a client that speaks MCP — in Claude, *Connectors* → *Add connector* → *Remote URL*; in the ChatGPT desktop app, *Settings* → *Plugins* → *add an MCP server* over streamable HTTP — and then asks for what they want in ordinary language. Nothing about the RDF shape or SPARQL syntax has to be learned first, which was the barrier the endpoint alone did not remove.

We recommend adding TogoMCP (<https://togomcp.rdfportal.org/mcp>) alongside it. The two together are what produced Figure 2, and the division of labour between them is the subject of the two sections above.

Ongoing work

MBEX, the MarpolBase expression database released in 2022, is being rebuilt: its content is growing from 340 to about 1,400 entries. The bottleneck there is not the data but the BioSample metadata that describes it, and we have built a tool that uses an LLM to assist that curation. The target is to have the expanded expression data in place for the international *Marchantia* workshop in Kobe in November 2026. Everything in this report therefore describes a resource that is still growing, and the 161 conditions used in Figure 1 are a snapshot of it.

Discussion

Publishing MarpolBase as RDF made it possible to link a species-specific plant genome resource to external resources, and registering the endpoint with TogoMCP and RDF Portal contributed a plant genome resource to the shared Japanese infrastructure. The most transferable lesson is the one in Table 2 and the section that follows it: for a resource covering a species that the major cross-reference hubs do not index, the integration point is not the gene identifier but the shared vocabularies the resource chooses to assign — in our case, above all, its own KO assignments.

The MCP server also changed how the resource is used in practice. Every analysis in this report was driven from a natural-language prompt rather than from hand-written SPARQL, and the value of the typed tools was less in saving keystrokes than in carrying the metadata along with the values — condition attributes with the expression numbers, evidence and role codes with the literature links — so that the agent could report *where* a gene is expressed and *on what grounds* a paper is attached to it, not only how much and how many.

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Author contributions

Yasuhiro Tanizawa: Conceptualization, Software, Writing – original draft; Takatomo Fujisawa: Software, Writing – review & editing.

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